PROMPT INJECTION DEFENSES FINAL REPORT *

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ABSTRACT

Prompt injection is a growing security risk for applications that use large language models (LLMs), especially systems that combine trusted instructions with untrusted user or retrieved text. To prevent Prompt Injection's many companies implement tools like Prompt Injection Scanners that will detect malicious content like this.

Although a great tool, it is not fully reliable when used alone. So we conducted a case study evaluating similar tools, where we observed distinct differences in Rebuff, Vigil, and Prompt Injection. Rebuff relies on a heuristic scoring mechanism that proves effective at flagging both obvious and borderline threats by evaluating risk against a defined threshold. In contrast, Vigil utilizes a YARA-based framework that offers high transparency and precision for established attack signatures, though it struggled with subtle inputs that fell outside its predefined rule set. Along with that, Prompt Injection focuses solely on detecting and filtering malicious input. Comparing these tools revealed that, when used alone, they can be considered weak, but working together they form a stronger prompt injection tool.

Thus, we developed a multi-layer detection pipeline that integrates a heuristic layer inspired by Rebuff with the machine learning classifier from PromptInjection. Our experimental results indicate that this ensemble approach significantly bolsters detection robustness by compensating for the specific limitations of each individual method. For instance, while the heuristic component efficiently identifies explicit keyword-based threats, it often overlooks nuanced prompts, that lack obvious indicators. The PromptInjection model, however, successfully flagged these instances by analyzing underlying intent rather than just surface patterns. By combining these two strategies, the integrated system achieves a broader detection coverage than any single tool, meeting our objective of enhancing and unifying existing frameworks into a more robust defensive solution.

***K*eywords** Prompt Injection Defenses · Rebuff Tool · Vigil Tool

* *Citation*: **Authors. Title. Pages.... DOI:000000/11111.**

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